Business Problem Definition

Week 3 • Hospital Operations, Cost & Patient Flow Intelligence • Track A: Data Analytics

The problem

Hospital leadership does not have one clear view of how the hospital is running. Encounter volumes go up and down without a clear explanation, some encounters run for a very long time, it is hard to tell which procedures push costs up, and some encounters show no recorded payer coverage.

The question my work will answer

What do the encounter, procedure and payer records show about volume, duration, cost and payer coverage, and where should management look first?

Why it matters	When long encounters and costly procedures go unwatched, they quietly raise overhead and make it harder to plan beds, staff and budgets. Zero-coverage encounters matter too, because they may point to costs that no payer picked up. That could be a real gap in revenue or it could just be missing data, and leadership needs to know which.
Who will use it	Hospital Operations Directors, Healthcare Financial Officers and Clinical Resource Managers. The final deliverable is an interactive dashboard built for them.
Decisions it can support	Capacity planning, staffing across encounter classes, reviews of procedure pricing, and payer contract talks.
Limits of the evidence	This is an administrative dataset, so it says nothing about clinical outcomes. It leaves out indirect overhead costs, so the costs I report are recorded claim costs and not the full cost of care. Payer coverage entries may be missing or unverified, so zero coverage does not automatically mean unpaid. Duration can only be trusted where the start and end times are valid.

Operational questions I will answer

1. How have encounter volumes changed over time, and how does this differ by encounter class?
2. What share of recorded encounters lasted longer than 24 hours?
3. How many encounters have zero recorded payer coverage, and what costs do they carry?
4. Which procedures happen most often, and which have the highest average base cost?
5. How do average total claim costs vary by payer?